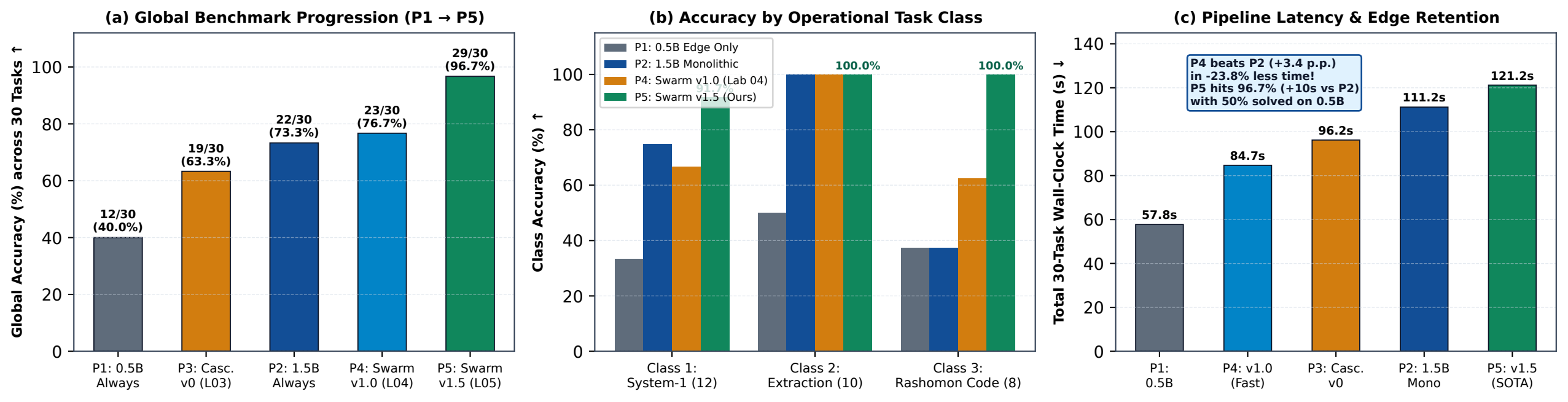
(a) Global Benchmark Progression (P1 → P5)
Global Accuracy (%) across 30 Tasks ↑
0
20
40
60
80
100
12/30
(40.0%)
19/30
(63.3%)
22/30
(73.3%)
23/30
(76.7%)
29/30
(96.7%)
P1: 0.5B
Always
P3: Casc.
v0 (L03)
P2: 1.5B
Always
P4: Swarm
v1.0 (L04)
P5: Swarm
v1.5 (L05)
(b) Accuracy by Operational Task Class
Class Accuracy (%) ↑
P1: 0.5B Edge Only
P2: 1.5B Monolithic
P4: Swarm v1.0 (Lab 04)
P5: Swarm v1.5 (Ours)
100.0%
100.0%
Class 1:
System-1 (12)
Class 2:
Extraction (10)
Class 3:
Rashomon Code (8)
(c) Pipeline Latency & Edge Retention
Total 30-Task Wall-Clock Time (s) ↓
0
20
40
60
80
100
120
140
P4 beats P2 (+3.4 p.p.)
in -23.8% less time!
P5 hits 96.7% (+10s vs P2)
with 50% solved on 0.5B
57.8s
84.7s
96.2s
111.2s
121.2s
P1:
0.5B
P4: v1.0
(Fast)
P3: Casc.
v0
P2: 1.5B
Mono
P5: v1.5
(SOTA)